JUEE NAIK

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SUMMARY

Data scientist with 3+ years building end-to-end ML pipelines on large-scale, high-dimensional time-series, imaging, and tabular data. Strengths in feature engineering, ensemble modeling, statistical validation in Python, SQL, and MATLAB.

EDUCATION

University of Pennsylvania (2024 – 2026): MSE in Bioengineering (Concentration in Neuroengineering), GPA: 3.84
Indian Institute of Technology (IIT) Madras, India (2020 – 2024): BTech in Engineering Physics, CGPA: 8.07

TECHNICAL SKILLS

- **Programming & Databases:** Python (pandas, NumPy, scikit-learn, seaborn, PyTorch, Tensorflow), SQL, MATLAB, Wolfram Mathematica, C++
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, YOLOv5, Faster R-CNN, Detectron2, XGBoost
- **Tools:** Git, WSL, LaTeX, MS Office, GraphPad Prism, Adobe Suite, AutoCAD

EXPERIENCE

Functional Connectivity Reliability Analysis during Episodic Memory Tasks *Jun '25 – Present*
Research Specialist - Computational Memory Lab, University of Pennsylvania

- Conducted comparative analyses on a large-scale iEEG event-matched time-series dataset (383 epilepsy patients), characterizing neural synchrony across multiple memory task phases (encoding, retrieval, distractor) and coupling between brain-wide synchrony and local spectral power
- Quantified reliability and test-retest reproducibility of 11 brain connectivity metrics through subsampling and statistical resampling methods
- Discovered low reliability and high FPR in detection of hypothesized memory-specific connections and network hubs at typical sample sizes
- **Results presented at CogNeuroSociety (CNS) - March 2026 and Context and Episodic Memory Symposium (CEMS) - May 2026**

Inter-brain Synchrony in Mice *Sep '24 – Present*
Graduate Researcher - Ma Lab, University of Pennsylvania

- Built automated behavior-classification pipelines using deep learning and HMMs (Detectron2, MATLAB) on multimodal experimental data
- Performed causal analysis on multivariate neural time-series in MATLAB to characterize cross-region connectivity in 5 frequency bands
- Quantified synchronization between socially interacting subjects using neural and behavioral datasets (OpenEphys, Boris)
- **Results presented at ACHEM-S XLVII (2025) and ACHEM-S XLVIII (2026) annual conferences**

Computational Modeling of Deep Brain Stimulation for Motor Disorders *Aug '23 – May '24*
Undergraduate Research Assistant - CNS Lab, IIT Madras, India

- Built an FEM solver (finite-difference Poisson, anisotropic conductivity from DTI tensors) to simulate DBS voltage spread
- Integrated simulation outputs with a biophysical spiking model to study network-level effects of stimulation
- Conducted parameter sweeps and sensitivity analyses across current, contact geometry, and tissue models as a virtual clinical testing ground
- Simulated 2 reward-learning tasks for Parkinson's Disease patients vs. healthy controls
- **Presented results at the Society for Neuroscience (SfN) Annual Conference, Oct 2024**

Wrist Movement Classification on EMG Signals *Jun '23 – Aug '23*
Research Intern - BMD Lab, Hokkaido University, Japan

- Built end-to-end ML pipeline (data acquisition, feature engineering, classification) for physiological signal classification
- Extracted time- and frequency-domain features from raw signal data in MATLAB and validated feature relevance through statistical analysis
- Trained and evaluated a Random Forest classifier, achieving 91% accuracy across 4 wrist movement classes
- **Published and presented results at the IEEE IATMSI-2024 Conference**

Comparison of Object Detection Models for Lung Cancer Nodule Identification *Apr '23 –Jun '23*
Data Science Intern - Periwinkle Technologies Pvt Ltd, Pune, India

- Trained and benchmarked YOLOv5 vs. Faster R-CNN on 1,000 lung CT images
- Faster R-CNN reached 89% mAP while YOLOv5 ran 6× faster at ~10× smaller size - tradeoff analysis informed deployment for low-resource screening

PROJECTS

ECoG-Based Finger Flexion Prediction Using Ensemble Learning *Spring 2025*

- Built a feature-driven ML pipeline to decode continuous finger flexion trajectories by turning raw human ECoG signals to predictions, across 3 subjects
- Engineered time-domain and high-gamma spectral features (spectral entropy, local motor potential, multi-band power) with lagged temporal windows, applied CAR, Butterworth, and notch filtering
- Stacked per-finger XGBoost ensembles via a RidgeCV meta-learner with adaptive Savitzky-Golay post-processing, achieved high Pearson correlation on held-out data

Real-Time EEG-to-Music Neurofeedback Pipeline *Spring 2026*
Signal Processing Lead - Penn Neurotech Society, University of Pennsylvania

- Led team in building a real-time pipeline translating scalp EEG signals into adaptive music selection from a Spotify playlist
- Implemented live artifact-robust processing and feature extraction computing alpha suppression (attention) and its rolling variability (arousal)
- Streamed live focus features to the hardware team's haptic wristband, closing the loop by delivering vibration cues to nudge the user

LEADERSHIP

Consultant (Early-stage MedTech Startup) *Penn Biotech Group, University of Pennsylvania*
Design Super-Coordinator | QMS Manager | Fine Arts Club Captain *IIT Madras*